

Immune effects of cocoa procyanidin oligomers on peripheral blood mononuclear cells.



There has been considerable work on the relationships between nutrition and the immune response, particularly on studies that have focused on adaptive responses. There is increasing recognition of the importance of innate immunity in host protection and initiation of cytokine networks. In this study, we examined the effect of select cocoa flavanols and procyanidins on innate responses in vitro. Peripheral blood mono-nuclear cells (PBMCs), as well as purified monocytes and CD4 and CD8 T cells, were isolated from healthy volunteers and cultured in the presence of cocoa flavanol fractions that differ from another by the degree of flavanol polymerization: short-chain flavanol fraction (SCFF), monomers to pentamers; and long-chain flavanol fraction (LCFF), hexamers to decamers. Parallel investigations were also done with highly purified flavanol monomers and procyanidin dimers. The isolated cells were then challenged with lipopolysaccharide (LPS) with quantitation of activation using CD69 and CD83 expression and analysis of secreted tumor necrosis factor (TNF)-alpha, interleukin (IL)-1beta, IL-6, IL-10, and granulocyte macrophage colony-stimulating factor (GM-CSF). The chain length of flavanol fractions had a significant effect on cytokine release from both unstimulated and LPS-stimulated PBMCs. For example, there was a striking increase of LPS-induced synthesis of IL-1beta, IL-6, IL-10, and TNF-alpha in the presence of LCFF. LCFF and SCFF, in the absence of LPS, stimulated the production of GM-CSF. In addition, LCFF and SCFF increased expression of the B cell markers CD69 and CD83. There were also unique differential responses in the mononuclear cell populations studied. We conclude that the oligomers are potent stimulators of both the innate immune system and early events in adaptive immunity.